- *Electronic warfare (EW)*: Certain drones act as signal jammers or decoys, while others gather intelligence, ensuring multi-role mission adaptability.
- *Search and rescue (SAR)*: Swarm UAVs can rapidly locate survivors or identify hazards in combat or disaster zones through cooperative area mapping.
- *Loyal wingman missions*: Swarms can accompany manned aircraft, sharing sensor data, extending radar coverage, or autonomously engaging hostile targets.

Notable examples include the US DARPA's Gremlins and OFFSET programs, China's swarming wingman projects, and India's ALFA-S swarm drone initiative, all designed to develop autonomous, cooperative UAV systems capable of functioning under GPS-denied and contested conditions.

B. Commercial and civilian applications:

Beyond defense, swarm technology has begun reshaping civilian sectors through its scalability, efficiency, and precision.

- *Agriculture*: Drone swarms perform synchronized crop monitoring, pesticide spraying, and soil analysis, reducing time and resource use.
- *Disaster management*: Swarms create real-time 3D maps, coordinate search operations, and deliver supplies to hard-to-reach areas.
- *Infrastructure inspection*: Coordinated drones inspect power lines, bridges, and pipelines simultaneously, providing comprehensive data in less time.
- *Logistics and delivery*: Swarm algorithms enable fleets of drones to manage package delivery routes collaboratively, improving efficiency and reducing congestion.